

CT214H – Web Programming

Practical Session 2 – CSS

Content

- ❖ Styling webpages with CSS:
 - Using CSS selectors to select webpage elements.
 - Using CSS properties to format the selected elements.

Ex 1. Do the examples in the lecture.

- ❖ **Objective:**
 - Learn how to use the CSS selectors.
 - Use CSS properties to format the webpage elements
- ❖ **Requirement:**
 - Do the examples in the lecture, browse them, and observe the results.
 - Modify the examples and observe the results.

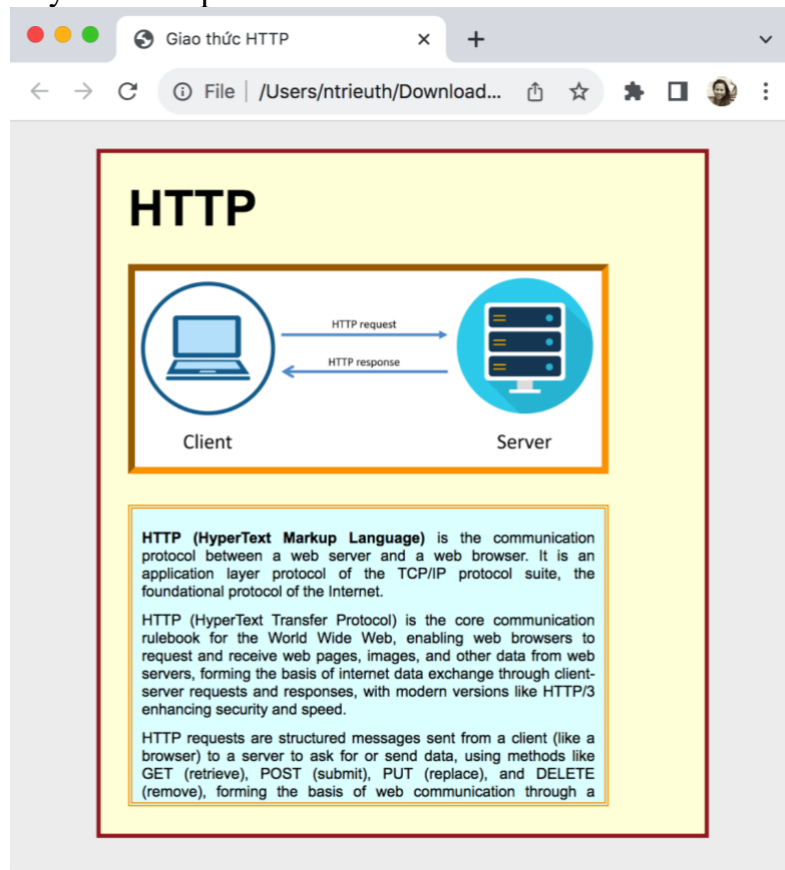


Figure 1. HTTP Protocol

Ex 2. Using the CSS box model.

- ❖ **Objective:** Using the CSS box model in designing webpages.
- ❖ **Requirement:** Create a webpage similar to Figure 1 with the following properties:
 - Outer box width: 400px
 - Image width: 340px
 - Space between the image and the text below: 20px

- Text window height: 200px.

❖ **Hints:**

- Use the <div> tag to group the elements.
 - The page contains 1 outer box (dark red border) with 1 heading, 1 image, and 1 inner box for displaying the text. Suggested steps:
1. Create the boxes and insert the content into these boxes. Ignore the format in this step.
 2. Format the boxes:
 - Outer box: width 400px, background color LightYellow, border attribute 3px solid brown. You can add padding between the content and the border.
 - Image border: style inset, width 5px, and Orange color, margin-bottom as 20px.
 - Inner box (containing text): height 200px, border-style as double, and background as LightCyan. Note: use the overflow attribute appropriately to avoid the content flowing out of the box.

Ex 3.

- ❖ **Objective:** Transform the un-styled HTML above into a beautiful, modern blog website by progressively adding CSS styles.
- ❖ **Preparation steps:**
 1. Download the HTML file: techblog.html
 2. Create a new file "style.css" in the same folder
 3. Link the CSS file in the HTML: <link rel="stylesheet" href="style.css">
- ❖ **Requirement:** step-by-step adding the style as describe in the tasks below and observe the results

1. Basic style reset and typography (using selectors, color, font, margin)

a. CSS Reset

- Remove default margins and padding from all elements using universal selector (set margin and padding to 0)
- Set box-sizing to border-box for all elements
- Set a default font family (Arial, sans-serif)

b. Body

- Set background color to #f5f5f5
- Set text color to #333
- Set line-height to 1.6

c. All headings (h1, h2, h3, h4)

- Set font-weight to bold
- Set appropriate font-sizes (h1: 2.5em, h2: 2em, h3: 1.5em, h4: 1.2em)
- Set color to #2c3e50
- Add margin-bottom 1em

d. Style paragraphs

- margin-bottom: 1em
- line-height: 1.8

e. Style links

- color: #3498db
- text-decoration: none
- Add hover effect (color: #2980b9)

2. Navigation styling: Fixed navigation bar, logo, menu styling (using flexbox, positioning, z-index)

a. Navigation bar

.navbar:

- background-color: #2c3e50
- padding: 1rem 0
- position: sticky
- top: 0
- z-index: 1000 (read more on https://www.w3schools.com/css/css_z-index.asp)
- box-shadow: 0 2px 5px rgba(0,0,0,0.1)

.nav-container:

- max-width: 1200px
- margin: 0 auto
- padding: 0 20px
- display: flex
- justify-content: space-between
- align-items: center

b. Logo

.logo:

- font-size: 1.5rem
- font-weight: bold
- color: #ecf0f1
- cursor: pointer

c. Navigation Menu

.nav-menu:

- display: flex
- list-style: none
- gap: 2rem
- margin: 0
- padding: 0

.nav-menu a:

- color: #ecf0f1
- font-size: 1rem
- transition: color 0.3s
- Add hover effect (color: #3498db)

3. Hero section: Gradient background, centered content

a. Hero Container

.hero:

- background: linear-gradient(135deg, #667eea 0%, #764ba2 100%)
- padding: 100px 20px
- text-align: center

- color: white

b. Hero Content

.hero-content h1:

- font-size: 3rem
- margin-bottom: 1rem
- color: white

.hero-subtitle:

- font-size: 1.3rem
- margin-bottom: 2rem
- opacity: 0.9

c. Hero Button

.hero-button:

- background-color: #fff
- color: #667eea
- padding: 15px 40px
- border: none
- border-radius: 25px
- font-size: 1rem
- cursor: pointer
- transition: transform 0.3s
- Add hover effect (transform: scale(1.15))

4. Container and layout: Grid layout

a. Main Container

.container:

- max-width: 1200px
- margin: 40px auto
- padding: 0 20px
- display: grid
- grid-template-columns: 2fr 1fr
- gap: 40px

b. Featured Section

.featured:

- grid-column: 1
- grid-row: 1

.section-title:

- font-size: 2rem
- margin-bottom: 30px
- padding-bottom: 10px
- border-bottom: 3px solid #3498db

c. Sidebar

.sidebar:

- background-color: #fff
- padding: 20px
- border-radius: 8px
- height: fit-content

5. Featured articles

a. *Featured Article Container*

.featured-article:

- background-color: #fff
- border-radius: 8px
- overflow: hidden
- box-shadow: 0 4px 6px rgba(0,0,0,0.1)

b. *Featured Image*

.featured-image:

- width: 100%
- height: 400px
- object-fit: cover

c. *Featured Content*

.featured-content:

- padding: 30px

.category:

- display: inline-block
- background-color: #3498db
- color: white
- padding: 5px 15px
- border-radius: 20px
- font-size: 0.85rem
- margin-bottom: 15px

.article-title:

- font-size: 2rem
- margin: 15px 0
- color: #2c3e50

d. *Article Meta*

.article-meta:

- display: flex
- gap: 20px
- color: #7f8c8d
- font-size: 0.9rem
- margin: 20px 0

6. Article grid: article cards in grid

a. *Articles Container*

.articles

- grid-column: 1
- grid-row: 2

.articles-grid:

- display: grid
- grid-template-columns: repeat(auto-fill, minmax(300px, 1fr))
- gap: 30px

b. *Article Card*

.article-card:

- background-color: #fff
- border-radius: 8px
- overflow: hidden
- box-shadow: 0 2px 4px rgba(0,0,0,0.1)
- transition: transform 0.3s, box-shadow 0.3s
- Add hover effect:
 - o transform: translateY(-5px)
 - o box-shadow: 0 8px 16px rgba(0,0,0,0.15)

.article-image:

- width: 100%
- height: 200px
- object-fit: cover

.article-body:

- padding: 20px

c. Card Content

.article-card .article-title:

- font-size: 1.3rem
- margin: 10px 0

.article-excerpt:

- color: #7f8c8d
- font-size: 0.95rem
- margin: 15px 0

.read-more:

- color: #3498db
- font-weight: bold
- display: inline-block
- margin-top: 15px

7. Sidebar widget

a. Widget Base Styling

.sidebar

- grid-column: 2;
- grid-row: 1 / 3;

.widget:

- background-color: #fff
- padding: 25px
- border-radius: 8px
- margin-bottom: 30px

.widget h3:

- font-size: 1.3rem
- margin-bottom: 20px
- padding-bottom: 10px
- border-bottom: 2px solid #ecf0f1

b. Newsletter Widget

.newsletter-form:

- display: flex
- flex-direction: column
- gap: 10px

.email-input:

- padding: 12px
- border: 1px solid #ddd
- border-radius: 4px
- font-size: 1rem

.subscribe-button:

- background-color: #3498db
- color: white
- padding: 12px
- border: none
- border-radius: 4px
- cursor: pointer
- transition: background-color 0.3s
- Add hover effect (background-color: #2980b9)

c. Lists Styling

.popular-list, .category-list:

- list-style: none
- padding: 0

.popular-list li, .category-list li:

- padding: 10px 0
- border-bottom: 1px solid #ecf0f1

.popular-list a, .category-list a:

- color: #555
- transition: color 0.3s
- Add hover effect (color: #3498db)

8. Footer

a. Footer Container

.footer:

- background-color: #2c3e50
- color: #ecf0f1
- padding: 40px 20px 20px

.footer-content:

- max-width: 1200px
- margin: 0 auto
- display: grid
- grid-template-columns: repeat(3, 1fr)
- gap: 40px
- margin-bottom: 30px

b. Footer Sections

.footer-section h4:

- color: #fff

- margin-bottom: 20px
- font-size: 1.2rem
- .footer-links:
 - list-style: none
 - padding: 0
- .footer-links li:
 - margin-bottom: 10px
- .footer-links a:
 - color: #ecf0f1
 - transition: color 0.3s
 - Add hover effect (color: #3498db)

d. Social Links & Footer Bottom

- .social-links:
 - display: flex
 - flex-direction: column
 - gap: 10px
- .footer-bottom:
 - text-align: center
 - padding-top: 20px
 - border-top: 1px solid #34495e

Ex4. CSS Layout Transformation

- ❖ **Objective:** Transform your TechBlog layout by moving the sidebar from RIGHT to LEFT
- ❖ **Preparation:** Create a new CSS file called layout-sidebar-left.css
- ❖ **Requirement:** step-by-step adding the style as describe in the tasks below and observe the results

1. Change container and layout

a. Main Container

- .container:
 - max-width: 1200px
 - margin: 40px auto
 - padding: 0 20px
 - display: grid
 - grid-template-columns: 1fr 2fr
 - gap: 40px

b. Featured Section on the right

- .featured:
 - grid-column: 2
 - grid-row: 1;

c. Sidebar on the left

- .sidebar:
 - grid-column: 1;
 - grid-row: 1 / 3;
 - background-color: #2c3e50

- padding: 30px
- border-radius: 8px
- height: fit-content
- border-right: 5px solid #3498db

2. Articles grid

a. Articles on the right

.articles:

- grid-column: 2
- grid-row: 2

b. Grid display

.articles-grid:

- display: grid
- grid-template-columns: repeat(2, 1fr)

3. Sidebar widget

a. Widget Base Styling

.sidebar .widget h3:

- font-size: 1.3rem
- margin-bottom: 20px
- padding-bottom: 10px
- border-bottom: 2px solid #3498db
- color: #3498db

.sidebar a:

- color: #ecf0f1
- transition: color 0.3s

.sidebar a:hover

- color: #3498db
- transition: color 0.3s

b. Newsletter Widget

.email-input:

- padding: 12px
- border: 1px solid #34495e
- background-color: #34495e
- color: white
- border-radius: 4px

c. Lists Styling

.sidebar p, .popular-list a, .category-list a

- color: #000
- transition: color 0.3s